

Grain size refinement of FePt granular films by introducing AlN-C and NbN-C as grain boundary materials for HAMR media

Kim Kong Tham¹, Ryosuke Kushibiki¹, Daiki Miyazaki¹, Shin Saito²

¹TANAKA PRECIOUS METAL TECHNOLOGIES Co., Ltd, Tsukuba 300-4247, Japan, tham@ml.tanaka.co.jp

²Department of Electronic Engineering, Graduate School of Engineering, Tohoku University, Sendai 980-8579, Japan

Investigation of nanostructure and magnetic properties of FePt-nitride-carbon granular films for heat assisted magnetic recording (HAMR) media with small grain size and high areal grain density is reported. Most of FePt granular films with co-addition of nitride and carbon show the grain size in between those of with pure-addition of nitride and carbon. Only granular films with co-addition of AlN-C and NbN-C show different characteristics with smaller grain size and higher areal grain density than expected even deposited at high substrate temperature of 550 °C. Especially the FePt-15vol%NbN-15vol%C granular film shows small average grain size of 3.8 nm and high areal grain density of 53.6 T/in². FePt-15vol%NbN-15vol%C granular film shows saturation magnetization of 542 emu/cm³ which is nearly equal to the volume average value of those of granular films with pure-addition of NbN and carbon, while perpendicular magnetic anisotropy of 3.2×10⁶ erg/cm³, which is significantly lower than the volume average value. Such a magnetic layer with soft magnetic properties and/or fine nanostructure is considered applicable for cap or buffer layers of HAMR media.

Index Terms— FePt-AlN-C, FePt-NbN-C, granular film, grain size, grain density, heat assisted magnetic recording.

I. INTRODUCTION

HEAT ASSISTED MAGNETIC RECORDING (HAMR) technology has been launched into the market as the technology to replace perpendicular magnetic recording. The L1₀ type FePt has been studied widely as a recording layer material for the HAMR media due to its high magneto-crystalline anisotropy of 5×10⁷ erg/cm³ at room temperature and moderate Currie temperature of 750 K. For the application of FePt thin film as the HAMR medium, it is necessary to resolve some essential issues, such as the formation of the granular structure with fine columnar magnetic grains, high areal grain density and perfect *c*-plane sheet texture. For the purpose of achieving these requirements, many studies of adding grain boundary materials (GBMs), such as carbon TiO₂ [3], Ta₂O₅ [4], GeO₂ [5], Al₂O₃ [6], Nb₂O₅ [7], Cr₂O₃ [8], SiO₂ [9], B₂O₃ [10], BN [11-12], TiN [13], AlN [14], NbN [14], and VN [14] into the FePt magnetic thin films while depositing at high substrate heating temperature to isolate the magnetic grains and reduce intergranular exchange coupling have been done. However, it was reported that when substrate heating temperature is increased, grain size of the granular film increases and areal grain density decreases significantly [15]. Previously, we have reported about the mixing of nitride with carbon as the GBMs of FePt granular films [16]. After carrying out additional investigations, we have found some peculiar nanostructure and magnetic properties for certain nitrides materials such as NbN and AlN when added with carbon that may shed light on the design of media with small grain size and high areal grain density. The detail investigation result will be reported in this paper.

II. EXPERIMENTS

All samples were prepared using RF/DC magnetron sputtering (Canon Anelva C3010-P7-UHV; 7-chamber configuration with single-side film fabrication, each process can be carried out in an independent chamber separated with

gate valve). The in-line sputtering equipment is almost the same model as that for the mass production of PMR media. The stacking structure of the samples is amorphous glass substrate (0.635 mm thick)/ Co₆₀W₄₀ (Ar gas pressure: 0.6 Pa, film thickness: 80 nm, substrate temperature: RT)/ MgO (4 Pa, 5 nm, RT)/ Magnetic layer (8 Pa, 5 nm, 550 °C)/C (0.6 Pa, 7 nm, RT). The magnetic layer was deposited at sputtering power of 0.5 W/cm². Fe₅₀Pt₅₀-15vol% nitride-15vol% carbon sintered targets were used for the magnetic layer. For comparison, the samples prepared with Fe₅₀Pt₅₀-30vol% nitride and Fe₅₀Pt₅₀-30vol% carbon sintered targets were also fabricated. For the nitride GBMs, Cr₂N, Si₃N₄, VN, AlN, NbN, TiN, BN, TaN and HfN are utilized. *M-H* loops were measured by using a vibrating sample magnetometer equipped with a SQUID detector (Quantum Design, MPMS3) at the maximum applied field of 70 kOe. Torque curves were measured by using a physical property measurement system with a torque magnetometer option (Quantum Design, PPMS) with the maximum applied field of 90 kOe [11]. Crystal structure was examined using out-of-plane and in-plane X-ray diffraction (XRD) measurements with Cu K α radiation at wavelength of 1.542 Å (Rigaku, SmartLab; 9 kW/ parallel beam configuration/ 2D detector). Nanostructure of the films was observed by transmission electron microscopy (TEM) with an acceleration voltage of 300 kV.

III. RESULTS AND DISCUSSION

Fig. 1 shows in-plane TEM images of FePt-nitride-carbon granular films with various nitride and carbon content ratios. For most of the GBMs in this experiment, when the content ratio of carbon in the GBMs increase, grain separation is promoted, grain size and areal grain density decrease with grain size and grain density in between those of with pure-addition of nitride (100:0) and carbon (0:100). Only granular films with co-addition of AlN-C and NbN-C show different characteristics, especially the FePt-15vol% NbN-15vol%C granular film shows

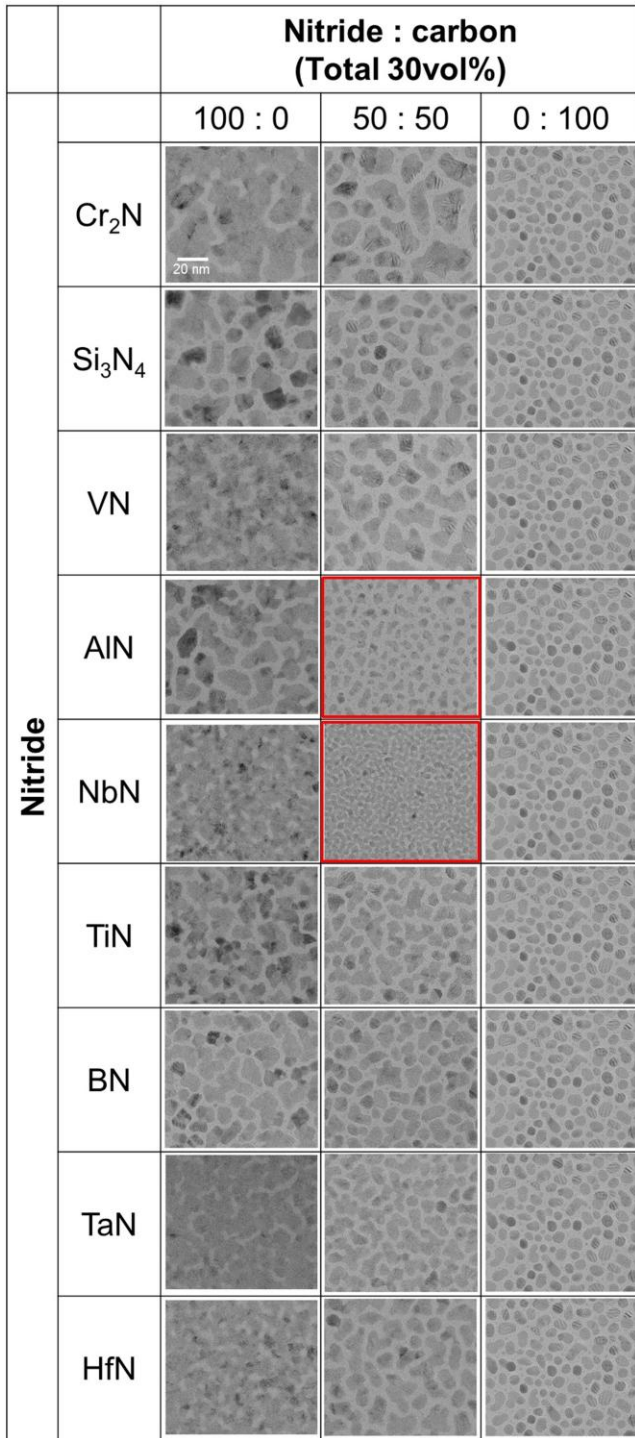


Fig. 1 In-plane TEM images of FePt-nitride-carbon granular films with various nitride and carbon content ratios.

small average grain size of 3.8 nm and high areal grain density of 53.6 Tera grains/in².

IV. CONCLUSION

According to the magnetic properties evaluation results, these films show relatively soft magnetic properties compared to that of FePt-30vol%C films. Therefore, for HAMR media, these

films are promising for a cap layer from the view point of soft magnetic properties and a seed layer from the view point of small grain size.

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